Course Project

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1. Course Project 1 / 1 point

Description. Cryptography is all about number theory, and all integer numbers (except 0 and 1) are made up of primes, so you deal with primes a lot in number theory. More specifically, some important cryptographic algorithms such as RSA critically depend on the fact that prime factorization of large numbers takes a long time. Basically you have a "public key" consisting of a product of two large primes used to encrypt a message, and a "secret key" consisting of those two primes used to decrypt the message. You can make the public key public, and everyone can use it to encrypt messages to you, but only you know the prime factors and can decrypt the messages. Everyone else would have to factor the number, which takes too long to be practical, given the current state of the art of number theory. Let us take a closer look at the "plain old RSA" algorithm.

For encryption, the RSA formula is as follows:

$F(m, e) = m^e \bmod n = c$

In this equation, m is the plain-text message, c is the resulting cipher-text message, and both e and n are the two components of the public key.

In this equation, the cipher-text c is the remainder of taking the plain-text m (expressed as an integer), and raising it to the power of e, and dividing by n using modular arithmetic.

For decryption, the RSA formula is as follows:

$F(c, d) = c^d \bmod n = m$

In this equation, c is the cipher-text, m is the resulting plain-text message, and both d and n are the two components of the private key.

In this equation, the plain-text m is the remainder of taking the cipher-text c (expressed as an integer), and raising it to the power of d, and dividing by n using modular arithmetic.

Example Encryption

Let the plain-text message m be the number 9.

For the two components of the public key, let e = 7 and n = 143

Step 1: Raise m to the power of e; $9^7 = 4782969$

Step 2: Find the remainder after dividing 4782969 by 143; 4782969 mod 143 = 48

The cipher-text message for m = 9 is c = 48

Example Decryption

Let us start with the cipher-text c = 48

For the two components of the private key, let d = 103 and n = 143 (same as before)

Step 1: Raise c to the power of d; $48^{103} = 1.4718 \times 10^{173}$

Step 2: Find the remainder after dividing 1.4718×10^{173} by 143; $1.4718 \times 10^{173} \bmod 143 = 9$

The cipher-text message for m = 9 is c = 48

Question: Using the same public key components e = 7 and n = 143, what is the cipher-text c if m = 15?

Correct

2. RSA Key Generation 1 / 1 point

Of course, the seeming "mathemagic" of RSA encryption and decryption depends on the RSA method for generating both the public and private keys. In "plain old RSA", key generation is a five-step process:

1. Choose two large prime numbers.

2. Calculate the Modulus using the two prime numbers.

3. Calculate the Totient of the Modulus.

4. Create the Public Key using the Totient.

5. Create the Private Key using the multiplicative inverse of the Public Key.

Question: Which of the following are prime numbers?

Correct

3. RSA Key Generation 1 / 1 point

Of course, the seeming "mathemagic" of RSA encryption and decryption depends on the RSA method for generating both the public and private keys. In "plain old RSA", key generation is a five-step process:

1. Choose two large prime numbers.

2. Calculate the Modulus using the two prime numbers.

3. Calculate the Totient of the Modulus.

4. Create the Public Key using the Totient.

5. Create the Private Key using the multiplicative inverse of the Public Key.

Question: If the Modulus is the product of two large prime numbers, which of the following results are the product of two prime numbers?

Correct

4. RSA Key Generation 1 / 1 point

Of course, the seeming "mathemagic" of RSA encryption and decryption depends on the RSA method for generating both the public and private keys. In "plain old RSA", key generation is a five-step process:

1. Choose two large prime numbers.

2. Calculate the Modulus using the two prime numbers.

3. Calculate the Totient of the Modulus.

4. Create the Public Key using the Totient.

5. Create the Private Key using the multiplicative inverse of the Public Key.

Question 4: Euler's Totient is the number of elements that have a multiplicative inverse in a set of modulo integers. In the RSA algorithm, the Totient of the Modulus is calculated as the product of the two prime factors, each minus one. In other words, given two prime factors, p and q, then the Totient of the Modulus, p * q, is calculated as (p – 1) * (q – 1). So, if p = 7 and q = 11, both prime numbers, then the Modulus = 7 * 11 = 77, and the Totient of the Modulus = (7 – 1) * (11 – 1) = 6 * 10 = 60, saying that in the sequence of integers between 1 and 77, sixty of those integers have a multiplicative inverse. If you understand this, then identify which of the following numbers is the Totient for the Modulus = 922?

Correct